Rthoptera Desk

User Manual

Desktop application for insect bioacoustics analysis

PROVISIONAL DRAFT — describes version 0.8.0

Rthoptera Desk project · October 2026

For colleagues' testing and feedback. Not for citation or redistribution.

Contents

1. Introduction and status of this manual
2. Installation
3. A tour of the window
4. Quick start: from a recording to a table
5. Loading audio and recording metadata
6. Preprocessing (Edit audio, Merge recordings)
7. Temporal Analysis
8. Spectral Analysis
9. Annotate
10. Plotting
11. Summarize and Summary Analysis
12. Preferences, language and text size
13. Output files and column names
14. Tips, troubleshooting and known limits
15. Citing the software and licence
Appendix A: Glossary of song structures
Appendix B: Feedback checklist for testers

1. Introduction

Rthoptera Desk is a cross-platform desktop application for analysing insect sounds (crickets, katydids, grasshoppers, cicadas and others). It is the graphical counterpart of the *Rthoptera* R package (Rivas et al. 2025): the temporal and spectral measurements and the acoustic metrics it reports are those defined in that paper. No R installation is needed; everything runs locally on your computer and no audio leaves it (the only features that use the internet are the map and elevation lookup in Recording metadata).

The workflow follows the structure of insect song. You load recordings, clean them if needed, detect and group sound units (*envelope peaks → pulses → motifs → motif sequences*), measure their timing and spectrum, make figures, and finally pool the results of many recordings into a species-level summary.

> **Status of this document.** This manual is a **provisional** draft written against version 0.8.0. The application is under active development: names of buttons, default values and some behaviours may change between releases, and some sections (marked *Provisional*) are shorter than they will eventually be. It contains no screenshots yet. Where the manual and the program disagree, trust the program and please tell the author. A checklist for reporting is in Appendix B.

Hovering the pointer over almost any control shows a tooltip explaining it; the information icon (ⓘ) next to some headings gives longer hints. Those tooltips are the most detailed reference for individual parameters; this manual explains how the pieces fit together.

2. Installation

Download the installer matching your system from the project's **Releases** page (github.com/panchorivasf/Rthoptera-Desk/releases).

System	File	Notes
Windows 64-bit	`Rthoptera.Desk_0.8.0_x64-setup.exe`	Recommended; double-click. The `.msi` is for managed deployment.
macOS, Apple Silicon (M1–M4)	`..._aarch64.dmg`	Apple menu → About This Mac shows "Apple M…".
macOS, Intel	`..._x64.dmg`	About This Mac shows an Intel processor.
Linux x86_64	`.AppImage` (run `chmod +x` first), `.deb`, `.rpm`	AppImage works on most distributions.

Windows-on-ARM and ARM Linux are not currently supported. Files ending in `.app.tar.gz` are for the auto-updater and are not meant for manual installation.

> Your operating system may warn about an unrecognised publisher the first time you open the installer or app. This is expected for software that is not commercially signed; confirm only if you downloaded it from the official Releases page. *(Provisional: signing status may change.)*

Supported audio formats: WAV, MP3, FLAC, OGG, AIFF. Use uncompressed WAV or FLAC for measurements whenever you can; lossy formats (MP3, OGG) alter the spectrum, especially at high frequencies.

3. A tour of the window

3.1 Layout

- **Top bar:** the **Preferences** menu (gear, left), the **Audio** button (load files) and the **Recording metadata** button.
- **Main tabs** (below the top bar): **Preprocessing**, **Temporal Analysis**, **Spectral Analysis**, **Annotate**, **Plotting**, **Summarize**.
- **Loaded Audio panel** (bottom of the window): a grid of thumbnails, one per loaded recording. The checkboxes here decide which recordings are used by batch edits, by the multi-recording plots and by Mean Spectra. **Select all** / **Clear selection** act on all entries; the gear sets the number of rows and columns.
- **Landing view:** before any audio is loaded you see a welcome page with a **Quick Guide** summarising the six steps.

3.2 Active recording vs. checked recordings

Two ideas recur throughout. The **active recording** is the single file that Preprocessing, Temporal, Spectral and Annotate currently work on (click its thumbnail to activate it). The **checked recordings** are those ticked in the Loaded Audio panel; they are used when an operation covers several files (Batch Edit, Osc. Stack, Multi-Mean Spectra, Habitus).

3.3 Common interactions

Action	Effect (most time-based plots)
Mouse wheel	Zoom in time
Ctrl + wheel	Zoom in frequency (spectrograms)
Shift + wheel	Zoom amplitude (Temporal Analysis) or pan (Auto-coded Oscillogram)
Drag	Pan or draw a selection, depending on the active tool
Shift/Ctrl + click or drag	Add to a selection / toggle items
Delete	Remove the selected detections or selections
Ctrl+Z	Undo (where an Undo button exists)
Ctrl+M	Merge selected detections (Temporal Analysis)
S / A / H	Select, Draw, Pan tools in Spectral Analysis (shown in tooltips)
Space	Play/pause in the Auto-coded Oscillogram

Hovering a tool shows the keyboard shortcut in its tooltip. In Temporal Analysis, holding `S` while the pointer is over the plot switches to split mode (cursor changes to a column-resize cursor).

4. Quick start: from a recording to a table

1. **Load** a recording with **Audio** (several files can be selected at once).
2. Open **Recording metadata** and fill in at least *Specimen ID* and *Species*; add temperature if you will compare recordings.
3. (Optional) In **Preprocessing** trim silence, high-pass filter below the insect's band, or downsample.
4. In **Temporal Analysis** press **Detect Envelope peaks**. Inspect the result; adjust *Smooth*, thresholds and gap parameters until pulses and motifs look right. Fix remaining errors by hand.
5. Press **Apply to Spectral Analysis**, switch to the **Spectral Analysis** tab and press **Spectral Metrics**.
6. Export with **Export All (Excel)** from each tab.
7. Repeat for the other recordings, then open **Summarize**, add all the Excel files and press **Merge & Summarize**.

> If your species has pulses that are poorly defined by envelope peaks (for example some grasshoppers), use the **Amplitude Detector** route instead of peak detection (section 7.5); it is faster to set up and can follow a changing noise floor.

5. Loading audio and recording metadata

5.1 Loading audio

Click **Audio** in the top bar (or **Load Audio** on the landing page). Selected files appear in the Loaded Audio panel. The **Audio Info** box on the Preprocessing tab reports duration, sample rate, Nyquist frequency and channels of the active recording.

Importing existing selections (Spectral Analysis tab, *Import selections*): **Import Selections** reads a Raven selection table (`.txt` / `.csv`); **Add Excel Selection Table** reads the Pulse or Motif table from a Temporal Analysis Excel export and turns it into selections.

5.2 Recording metadata

The **Recording metadata** window has two parts and is applied to the **active recording** as you type. Everything entered is written into the tables you export.

Section	Fields
Recording session	Country, Locality, Date, Time, Temperature (°C), Elevation (m), Latitude and Longitude (optional), and equipment: Recorder, Microphone, Accessories.
Specimen	**Specimen ID** (carried as the first column of every export, and used by Summarize to count individuals correctly) and **Species**.

- **Temperature** matters: stridulation rate depends strongly on it. Free text is accepted (22.5 or ~23).
- **Extract from map** opens a Leaflet map (street or satellite). Click a point, then **Use this point** to fill latitude, longitude and elevation. Elevation is also looked up automatically if you type coordinates and leave it empty. *Needs internet.*
- **Remember** stores the current values for reuse; the drop-down fills the fields from a remembered entry and the bin forgets it.
- **Apply session to all loaded recordings** copies the session block to every recording in the list (useful for a field night with many files).
- **Save to file / Load from file** writes/reads both sections as one `.json` (older `specimen_metadata.json` files still load).

5.3 Xeno-canto upload helper

Inside the metadata window, **Xeno-canto upload helper** lists, in order, the fields of xeno-canto's two-page upload form, each with a copy button; fields that are radio buttons or drop-downs on the site are shown as reminders. It does **not** send anything: you open the upload page from the helper and paste values yourself. Text you type in the helper is remembered on this computer.

Exported columns from metadata: `rec_date` , `rec_time` , `latitude` , `longitude` , `elevation_m` , `temp_c` , `recorder` , `microphone` , `accessories` , plus specimen ID and species.

6. Preprocessing

Two sub-tabs: **Edit audio** and **Merge recordings**. Edits made on the left apply to the **active recording** and are seen immediately by Temporal Analysis, Spectral Analysis and Annotate.

6.1 Edit audio (single recording)

Tool	What it does
Time selection	Click the waveform to place the playhead; drag to select a stretch. `|<` and `>|` jump between edges. A *Duration* box resizes the selection from its start (handy for cutting clips of equal length). **Save selection** writes the selection as its own WAV (original name + _1, _2, ...).
Trim	**Trim...**, drag the two handles (on the waveform or on the Spectral Analysis tab), then **Confirm Trim**. **Reset** restores the full duration. **Save Active Selection** exports the region without applying the trim.
Bandpass filter	High-pass and low-pass cut-offs in Hz (0 = off for high-pass; low-pass defaults to Nyquist). Use a high-pass just below the insect's lowest frequency to remove wind and handling noise.
Frequency drop	Scales every frequency down by a percentage **keeping duration** (25% moves 40 kHz to 30 kHz). A way to bring ultrasonic song into a convenient range.
Downsample	Lowers the sampling rate (an anti-aliasing filter runs first). Only lower rates are offered. Everything above the new Nyquist is lost permanently.
Undo Edit	Steps back the last edit.
Save Edited Audio As...	Offers a suffix reflecting the edits (e.g. `_1hpf_trimmed`). **Save (in app)** adds a new entry to Loaded Audio only; **Save & Export to Disk** also writes a WAV. The original is never modified.

6.2 Batch Edit

Tick recordings in the Loaded Audio panel (or use **All** / **None**) and apply: **Filter selected** (HP/LP), **Normalize selected** (target peak level in dBFS), **Drop selected** (frequency drop), **Downsample selected**. A checked recording already at or below the chosen rate is left untouched.

> Batch edits act directly and **in place** on the checked entries and cannot be undone; re-import the file to start over. **Save edited files...** writes each checked entry to a folder you choose as `<name>_<suffix>.wav`.

6.3 Merge recordings

Joins several recordings end to end (the equivalent of `merge_waves()` in the R package).

1. Build the list: **Add** ▸ from Loaded Audio. The list order is the playing order (top first).
2. If the sampling rates differ, a warning explains why concatenating as-is would be wrong (the odd ones would play at the wrong rate). Choose *resample to the highest / lowest / first*, or *must already match*.
3. Set *Gap between* (ms of silence inserted between pieces only), and optionally *centre* (subtract the mean) and *normalize to* a dBFS peak.
4. Name the result and press **Merge**. Join points are shown as dashed lines. The merged recording becomes the active one.

5. **Export WAV** writes a 16-bit mono WAV; **Export piece boundaries** writes a Raven table of where each source landed.

> Normalizing scales the whole concatenation to one peak, so amplitudes are no longer comparable between the source recordings. Turn it off if that matters. Unlike the R version, normalization is applied once at the end.

7. Temporal Analysis

This tab detects and groups the sound units of the song and measures their timing and amplitude. It works on the amplitude envelope of the active recording.

7.1 The song hierarchy used by the program

Level	Meaning	Defined by
Envelope peak	A local maximum of the smoothed envelope (e.g. one tooth impact).	Smooth, peak window, thresholds
Pulse	A group of envelope peaks close together (e.g. one wing stroke).	Max envelope-peak gap, max amplitude difference, min peaks per pulse
Motif (echeme / syllable group)	A group of pulses separated by short gaps.	Max pulse gap
Motif sequence	A group of motifs.	Max motif gap

Results appear in five sub-tabs: **Envelope peaks**, **Pulses**, **Motifs**, **Motif Sequences**, **Summary**.

7.2 The plot

The main canvas shows the envelope with detections drawn as semi-transparent fills above the line. Mouse wheel zooms time, Shift+wheel zooms amplitude (or use *Zoom X*, *Amp Y* and **Fit**), and drag pans. The overview strip below lets you click or drag to navigate. Tick **Spectrogram** to show a small synchronised spectrogram under the envelope, which helps distinguish real impacts from noise. Motif and sequence lines are drawn live above the pulses.

7.3 Detecting envelope peaks (route A)

1. Press **Detect Envelope peaks** (the first time, with defaults).
2. Tune the parameters below. **After the first Detect, changing any detection parameter re-runs detection automatically** after a short pause. Note that re-detection replaces manual peak edits.

Envelope peak detection

Parameter	Role
Smooth (ms)	Smoothing window of the envelope. Larger values merge nearby bumps.
Envelope peak window (ms)	Window used to decide that a point is a local maximum.
Envelope peak threshold (%)	Minimum prominence of a peak.
Detection thr. (%) and **Auto**	Minimum level for a peak to count. *Auto* sets it from the background noise (median + 3 spread of the quiet part of the envelope, below 25% of the peak). Typing a value switches Auto off.

Onset thr. (%)	Optional lower floor: weak peaks above it are kept only if they sit within one max peak gap of an accepted strong peak. Catches quiet onsets/offsets without picking up isolated noise.

Grouping

Parameter	Role
Max envelope peak gap (ms)	Largest gap between peaks inside one pulse.
Max amp diff (%)	Largest amplitude difference between neighbouring peaks of one pulse.
Min envelope peaks/pulse	Pulses with fewer peaks are dropped (and their peaks left unassigned).
Edge pad (ms/side)	Expands every pulse on each side to capture the true onset/offset; pulse rate and peak-timing metrics are unaffected.
Max pulse gap (ms)	Pulses closer than this belong to the same motif.
Max motif gap (ms)	Motifs closer than this belong to the same motif sequence.

Arch detection

For species whose pulses run back to back, so that the silence between pulses is no longer than the spacing of peaks within one, gap rules cannot separate pulses. Tick **Split on arches** to cut at the valley between amplitude arches. *Valley depth (%)* sets how deep a valley must be relative to the crests (lower = more splits); *Min pulse (ms)* refuses cuts that would make a pulse too short (blank = derived automatically). It runs after the gap rule and only subdivides.

Spectral parameters

Target frequency resolutions (in Hz, so identical across sampling rates) for spectra of envelope peaks, pulses and motifs. They do not affect detection or grouping. **Keep them constant across a study.** Where a unit is too short for the requested resolution, the coarser resolution actually achieved is reported in `spec_res_hz`.

Learn from edits

Correct a few pulses by hand, select their peaks (Shift-drag), and press **Fit to selection**: the program searches the grouping parameters that best reproduce your boundaries. Tick *incl. envelope peak detection* to fit the detection parameters too (slower; Apply then re-runs detection over the whole recording, replacing manual edits). Nothing changes until **Apply**.

7.4 Editing by hand

In the **Edit Mode** box choose a tool:

- **Select**: click a marker to select it; Shift-drag selects several.
- **Add Envelope peak**: click to add a peak snapped to the envelope.
- **Draw Selections**: draw your own temporal selection, independent of peak detection, and assign its *type* (Envelope peak, Pulse (train), Echeme (motif), Motif sequence, Other). Pulse-type draws feed the **Manual Echemes** table. **Clear draws** removes all drawn selections.

Envelope peak Actions (apply to the selected peaks): Assign left/right, Merge left/right, Split left/right, Isolate (make a single pulse), Join, Remove, Clear selection. **Reset boundaries** re-runs the segmentation and discards boundary edits (added/removed peaks are kept). **Drop unassigned peaks** deletes peaks that belong to no retained pulse. **Clear peaks** removes peaks but keeps the envelope on screen.

7.5 Amplitude Detector (route B)

Thresholds the envelope directly, with no peaks involved. Parameters: *Floor %*, *Min dur.* (ms), *Min gap* (ms). Press **Amplitude Detector**; detections can be edited by dragging their edges, merged (`Ctrl+M`) and deleted.

Local noise floor

Detect local minima finds the lowest points of the envelope within a window (*Floor win.*, ms; choose a window longer than the spacing between pulses), joins and smooths them, and draws a bright green curve. Afterwards, *Floor %* becomes an offset **above this curve**, so reverberation under loud passages no longer merges elements while faint parts are still caught. The ✕ button removes the curve and returns to a fixed level. Changing the window rebuilds the curve.

7.6 Computing metrics

Button	Use when
Compute Metrics from Peaks	You used peak detection (route A). Gives full per-peak, pulse and motif columns.
Compute Metrics from Pulses	You have amplitude detections or drawn/imported selections of type Pulse. Pulses are grouped into motifs by *Max pulse gap*. Dynamic excursion and maximum amplitude come from the envelope sampled every 1 ms inside each pulse; **temporal excursion and per-peak columns are not calculated**.

7.7 Importing earlier work

- **Import Peaks**: loads the Envelope peaks sheet of an earlier export and restores peaks and pulse boundaries exactly (including hand edits) without re-running detection. Needs the matching audio.
- **Import Pulses**: loads the Pulses sheet (`pulse_start` , `pulse_end`) as Pulse selections; then use *Compute Metrics from Pulses*.

7.8 Parameter presets

Parameter Presets offers a built-in **Acrididae** preset (Smooth 50 ms, Floor win. 50 ms) and ten user slots stored on this machine (**Save**, **Load**, **Delete**). **Export file…** / **Import file…** use a `.json` that can be copied to another computer. Presets include the Amplitude Detector settings, and loading one recomputes the envelope immediately.

7.9 Sending results on, and exporting

- **Apply to Spectral Analysis**: sends pulses or motifs (*Apply as*) as selections to the Spectral tab, however they were found. If some detections are selected only those are sent; a repeat Apply replaces the previous one.

- **Apply motifs to Annotate**: sends one selection per motif, labelled with the recording's Species and delimited in frequency by the power-spectrum threshold.
- **Export CSV** (current table) and **Export All (Excel)**: one workbook with sheets Envelope peaks, Pulses, Motifs, MotifSeqs, Summary.
- **Export Temporal Selections** (Manual Echemes): amplitude detections and manual annotations as one chronologically numbered list, with echeme statistics (TE/DE/PCI) from any Pulse (train) annotations.

Short notices appear at the bottom-right of the window for problems such as *no peaks detected*, *no pulses with enough peaks* or a failed import.

8. Spectral Analysis

Shows the waveform and spectrogram of the active recording, lets you create and edit selections, and measures their frequency content.

8.1 Display

- **Playback bar:** Play/Stop, jump between selection edges, **Speed** (1–400% of real time; slowing lowers the pitch, bringing ultrasonic song into hearing range — 1% plays a 120 kHz carrier at about 1.2 kHz, but takes 100× as long), and **Freq drop** (lowers pitch for listening only *without* stretching time). Neither changes the stored audio, measurements or exports.
- **Drop-down menus:** *Spectrogram* (contrast in dB, brightness, FFT size 128–8192, colour map: Inferno, Viridis, Hot, Cyan-Magenta, Gray), *Grid*, *Envelope & Floor* (RMS window, Floor %), *Show* (envelope, spectrogram, waveform, selections, detections).
- Waveform and spectrogram heights are adjustable by dragging their handles. Scroll zooms time; Ctrl+scroll zooms frequency.

8.2 Selections

The collapsible **Detection & selection tools** row has: **Select** (S), **Draw Selections** (A), **Pan** (H), *Temporal only* (ignore vertical drag, so selections cover the whole frequency axis), **Undo**, **Selections** (download a Raven table), **Clear Spectrogram**, plus the import buttons (section 5.1) and the selections list. Selections created elsewhere (Temporal Analysis) arrive here as well.

8.3 Measuring

1. Set *Measure res. (Hz)*: the frequency resolution used for measurements, independent of the display FFT. The program picks the next power-of-two FFT that reaches at least this resolution.
2. Press **Spectral Metrics**. A table of per-selection (or per-detection) values appears under **Detections & Measurements**, with summary statistics.
3. **Export All (Excel)** saves the workbook; **Export Text Report** saves a plain-language narrative (DOCX); **Explain Metrics** shows a description of every column (also downloadable as DOCX); **Clear Measurements** empties the table.

Export detections writes the amplitude detections and manual selections to a workbook. **Apply motifs to Annotate** is also available here.

> Which frequency features and how they are defined (peak frequency, bandwidth, centroid, entropy, flatness, etc.) is documented in the Rthoptera paper and in the in-app *Explain Metrics* window. *(Provisional: a complete column dictionary will be added to this manual.)*

9. Annotate

Annotate labels the motifs of **one species at a time** with a time span and a frequency band, then exports them to Raven or Xeno-canto formats. The idea: mark the time span on the waveform, set the frequency band on the power spectrum of that span, and add a selection.

9.1 Defining a selection

1. Type the **Species** (typed once, then offered from a list).
2. On the waveform, drag to set the time span (Shift+drag pans, wheel zooms, a click clears it).
3. On the power spectrum, drag horizontally to set the band, or let it be derived automatically: *Search Hz* (set it above wind/handling rumble, or the band will collapse towards DC), *Threshold* (band edges this many dB below the spectrum peak; 20 dB is the usual convention), and *edges*: **outer** (outermost crossings, harmonics included) or **around peak** (one lobe around the carrier). *auto* re-derives the band when the selection changes; *lock* freezes it for every later selection and recording. Type exact edges in *Band Hz* and press Apply.
4. Press **Add selection** (or Enter / A). Other controls: Undo, Clear span (Esc), Delete (Del), Zoom to span.

The spectrogram shows boxes but is not where they are drawn. Click a box to select it and adopt its label and band; drag a box's left/right edge to change its start or end. The *Spectrum* window size is deliberately not the finest available: the band should describe the species, not one individual or microphone.

9.2 Saving bands

Save band writes a `.band.json` (refused when the band was derived from the whole view, since in a recording holding two species that is neither species' band). **Load band…** adopts a saved band, locked.

9.3 Detect motifs in the band

With a band defined, **Detect in view** or **Detect in span** finds motifs in the band-limited envelope, so a second species elsewhere in the spectrum cannot fuse its sounds into these motifs. Parameters: *Smooth ms*, *Threshold %* (relative to the loudest moment in the stretch searched), *Gap ms* (silence that splits motifs), *Min dur ms*. Re-running skips any motif already more than half covered by an existing selection of the same species.

9.4 Managing selections

- **Rename a species** (also how selections that arrived from Spectral Analysis or a Raven import get a real name) — *Rename all*.
- In the **Selections** list: check all, **Delete checked**, **Relabel checked** (set the species of checked items), **Apply band** (set the current band on checked items).

9.5 Export

Choose a format: **Raven selection table (.txt)**, **Raven + Begin File / File Offset**, **Xeno-canto annotation set (.json)**, or **Generic CSV (s / Hz)**. For Xeno-canto, enter the recording number (digits

only; filled from a file name containing `XC...`), the annotator name, and optional fields applied to every annotation. Tick *checked only* to export a subset. Defaults come from Preferences → Default metadata.

10. Plotting

Six sub-tabs produce publication figures. Most have **Draw**/**Render** plus **SVG** (vector, editable in Illustrator or Inkscape; what most journals prefer) and **PNG** downloads. Presets save layout settings.

10.1 Multiplot

Waveform + spectrogram + power spectrum for one time/frequency window. Set Tmin/Tmax and Fmin/Fmax (**↔ view** and **&updown; view** copy the current analyser view), time unit, tick steps, colormap, contrast, brightness, scale (dB or linear), figure width/height/DPI (72–288), title and font. Sections: *Figure Layout* (spectrogram/waveform ratio, power-spectrum width, noise floor), *FFT* (auto window length or override 256–16384, overlap 75/87.5/93.75%), *Colors*, *Power Spectrum*, *Grid*, *Typography* and a *Parameters stamp*. Press **Render**, then **PNG**. Presets: 10 slots plus **Export**/**Import** of a `.json` file.

10.2 Osc. Stack

Stacks the oscillograms of several recordings (tick them in Loaded Audio, then **Add selected**). Controls: axis buffer and time axis (defaults to the longest wave plus the buffer), scale bar, trace colour/width, row height, trace labels, per-wave normalization. Extras: **Add letters** (panel letters), **Add annotation**, symbols (♂/♀). Click a label to select it (Shift/Ctrl for several, or rectangle-drag); drag or use arrow keys to move; double-click to edit text. A *Selected label style* box sets font, size, thickness and colour.

10.3 Osc. Zoom

Shows one wave with zoomed panels. Drag across any panel to open that slice below it (up to *Maximum panels*). Options: panel height, *Context padding (%)* (extra audio on each side of a selection), time reference (scale bar or axis), scale-bar length per panel (**Set**, **Auto**, *All panels*), and *Rescale each panel to its own peak*. **Reset zoom** clears the panels. Presets store sidebar settings only; panel ranges and label text belong to the recording.

10.4 Auto-coded Oscillogram (CETPE)

An oscillogram whose colour marks where chosen frequency bands rise above a pressure threshold (after Brizio 2023). **Please cite Brizio (2023)** for any figure made here (a **Copy citation** button is provided).

1. **Analysis:** FFT size (time resolution is *nfft/fs*), overlap, and *Rendering*: 2-colour (one contrasting colour) or Multicolour with 6/12/24 bands (each band mapped to one bit of the 24-bit RGB triplet). Press **Analyze**.
2. **Frequency window:** FRB and FRT (Frequency Range Bottom/Top, Hz); **Full recorded band** resets. **FRPT** (dBFS) is the pressure threshold at which a band counts as present; changing it recolours instantly. Changes to FRB/FRT/FFT need a new Analyze.
3. **Pressure range (optional):** restrict colour to columns whose overall level lies between PRB and PRT.
4. **Figure:** toggle the reference oscillogram, reference spectrogram, band→colour legend and parameter caption; set heights, colours and spectrogram options.
5. **Redraw**; export **SVG**/**PNG** (with a scale factor) or **Coloured regions (CSV)**, a table of contiguous coloured intervals with the bands that lit them.

On the figure: drag to zoom, click to place the playhead, wheel to zoom, Shift+wheel to pan, double-click or `Home` for the whole recording, `Space` to play. A live indicator lights when the playhead is over a highlighted column. Presets can be exported/imported as `.json`.

10.5 Habitus

Combines a photo of the animal with a waveform, spectrum and labelled selections: pick the wave, window length and opacity, drag on the oscillogram to brush a stretch, name it (closing, opening, male, female or custom) and **Add selection**. Export PNG, SVG or the spectra as CSV; set a file prefix. *Provisional: this module is documented only briefly.*

10.6 Multi-Mean Spectra

Tick the recordings to average in Loaded Audio (no separate add step). Spectrum options: window length (128–8192), overlap, window function (hanning, hamming, blackman, bartlett, flattop, rectangle), scale (linear, dB, dBFS), frequency limits and step, minimum amplitude, and *Normalize each recording before averaging*. **Contour + Band** draws the central statistic (mean or median) with a band of ± N×SD in the colours and opacity you choose. An optional habitus photo can be added and the A/B letters dragged. All recordings must share the same sampling rate. Exports: PNG, SVG, spectra CSV.

11. Summarize and Summary Analysis

This tab needs no audio. It pools the Excel exports of many recordings into species-level statistics.

11.1 Summary sub-tab

1. **Source files:** Add Excel Files to load the Temporal and/or Spectral Analysis `.xlsx` exports of each recording. The Specimen ID tagged at export is read automatically; if missing or wrong, edit it in the list: this is what groups recordings by animal. **Clear all** empties the list.
2. (Optional) **Temperature regression:** enter a reference temperature in °C. Every metric is regressed on the temperatures actually recorded (least squares) and expressed at the reference temperature; adds a corrected mean/SD beside each observed one, a bracketed figure in the report and Temp_Regression sheets in the workbook.
3. **Range brackets:** choose `( )` or `[ ]` for min–max in the LaTeX/Word convenience columns (they are Excel formulas, so change and save again to switch).
4. **Structure selections** (optional): summarize a named subset of structures separately (first pulse of every echeme, odd-numbered strokes, last pulse, …). Positions are counted inside each parent structure, per recording. Gap levels (e.g. 1-2, 3-4, or *odd/even* on "Gaps between pulses within each echeme") select silences; run levels treat several structures as one sound (a disyllabic species is "Syllables (runs of pulses)" with runs of 2, measured from first onset to last offset). Runs must be complete; leftovers are reported as left over. Each selection gets its own rows, sheets and report paragraph. With none defined, everything is summarized together.
5. Press **Merge & Summarize**. Outputs: *Pooled summary* (counts and mean ± SD per metric), *Summary table (by specimen)*, *Temperature response* plots (SVG, PNG 300 dpi) and a *Text report*.
6. **Save Merged Workbook** writes the merged `.xlsx`; **Save Text Report (DOCX)** writes a draft narrative.

> Older workbooks: the `props_ent`, `props_cv` and `pci` columns of the Manual Echemes sheet are automatically renamed `props_ent_syl`, `props_cv_syl`, `pci_syl` on import.

11.2 Summary Analysis sub-tab (compare species)

Run **Merge & Summarize** first, then open **Summary Analysis**.

Variables and species

- **One observation per** recording or specimen; each observation is the mean of every chosen variable over its rows.
- **Tables** (Motifs, Pulses, EnvPeaks, Spectral, MotifSeqs): where variables come from. The default is the motif level, because the same measurement averaged at several levels is nearly the same number repeated, which over-weights it. Add Pulses or EnvPeaks for things the motif row lacks, such as pulse gaps.
- **Drop near-duplicates** (correlated above a threshold): of each group of correlated variables keeps the one that separates species best. **Suggested set**, **All**, **None** shortcuts.
- **Species shown** (All / First 6 / None) and **Names and labels** to rename recordings or species.

- **Adjust for temperature to** a reference °C, within species, optionally only for variables with significant slope ($p < 0.05$). The variable ranking has a Temp. R^2 column.
- **Reload from merged data**, **Reset analysis**, **Save settings** / **Load settings…** (`.json`).

Analyses

Output	What it shows
Run analysis	Computes everything below. *Scores, loadings & relevance (CSV)* exports the numbers.
PCA biplot	Choose X/Y components; optional 1σ/1.5σ/2σ ellipses and loadings. PNG/SVG.
3D PCA	PC1–PC3 with 1.5σ ellipsoid outlines (needs 3+ variables). Hover/click for recording IDs; export PNG or an interactive HTML page (works offline).
Radar profiles	Each variable scaled 0–1; line = species mean, band = ±1 SD.
Which variables separate the species?	η^2 (share of variance explained by species, one-way ANOVA) and F per variable. One variable at a time, so correlated variables rank similarly.
Compare recordings or species	Pick 2–6 items for a side-by-side radar and a table of real values and ratios.
Leave-one-out test	Classifies each observation from all the others using Mahalanobis and LDA; leaving out the whole specimen is the stricter, more honest test.
Species profiles and knowledge base	(1) *Extract species profiles* (mean and spread per species), (2) save to / load a **knowledge base** file, (3) *Compare with knowledge base*: place a new song among known species and see the closest by LDA or Mahalanobis.

> *Provisional.* Summary Analysis is new (0.7.3) and still being refined. Treat classification accuracy from small samples with caution, and avoid pseudo-replication: several recordings of one animal are not independent observations.

12. Preferences, language and text size

The **Preferences** menu (gear, top left) contains:

- **Home:** returns to the landing view.
- **Default metadata…:** values you reuse. *Country* and the *default equipment* (recorder, microphone, accessories, one per line with a default for each) are filled into every newly loaded recording; *Recordist name* and *License* are used by the Xeno-canto helper; *Annotation author* is used by the Annotate export. Equipment typed in a recording's metadata joins these lists automatically. Everything is stored on this computer.
- **Language:** English, Español, Italiano, Português.
- **Text Size:** Small (90%), Normal, Large (115%), Extra large (130%).
- **Version:** shows the installed version (please include it in bug reports).

Presets, remembered metadata and preferences are stored locally by the app. They are not part of your recordings or exports, so copy preset `.json` files if you want to move them between computers.

13. Output files and column names

Where	File	Contents
Temporal Analysis	Excel workbook	Sheets Envelope peaks, Pulses, Motifs, MotifSeqs, Summary (plus Manual echemes and Info where applicable)
Spectral Analysis	Excel workbook, DOCX report	Per-selection spectral metrics, summary statistics, narrative
Annotate	`.txt`, `.json`, `.csv`, `.band.json`	Raven, Xeno-canto and generic selections; saved bands
Plotting	SVG, PNG, CSV, `.json`	Figures, coloured-region intervals, mean spectra, presets
Summarize	Excel workbook, DOCX, CSV, HTML, PNG/SVG	Merged workbook with stats and temperature sheets, text report, PCA outputs, interactive 3D plot, knowledge base

Every table starts with the Specimen ID and carries the recording metadata columns listed in section 5.3. Columns containing `_pulse_mean` in the motif table are averages of that motif's pulse rows (which contain no silence); compare them with the whole-motif spectral values, which include the gaps and therefore tend to raise entropy and flatness on low duty-cycle songs. The pulse-based complexity columns are `pci_syl`, `props_ent_syl` and `props_cv_syl` in both the Motifs and Manual Echemes tables.

> Scripts of your own that read the old Manual Echemes names (`props_ent`, `props_cv`, `pci`) need updating.

Provisional: a full column dictionary will be added. Until then use Explain Metrics in Spectral Analysis and the Rthoptera paper.

14. Tips, troubleshooting and known limits

14.1 Good practice

- Keep spectral-resolution parameters, smoothing and presets **identical** across all recordings of a study.
- Always record temperature and tag Specimen ID and Species before exporting; fixing them afterwards means re-exporting.
- Work on a copy of your recordings; edits never overwrite the originals, but Batch Edit cannot be undone inside the app.
- Save presets as `.json` files next to the recordings they were used for, so the analysis can be reproduced.
- Check detection visually in every recording. Automatic thresholds are a starting point, not a guarantee.

14.2 Common problems

Symptom	Try
No peaks detected / notice at bottom right	Lower Detection thr. or tick Auto; reduce Smooth; check you are on the right recording; try the Amplitude Detector.
Pulses merged together	Lower Max envelope peak gap; for back-to-back pulses tick Split on arches; with the Amplitude Detector use Detect local minima and a smaller Min gap.
Pulses split too much	Raise Max envelope peak gap or Max amp diff; raise Valley depth; raise Smooth.
My manual edits disappeared	Changing a detection parameter, pressing Detect again, or Apply in Learn from Edits re-derives the peaks and replaces manual edits. Edit last.
Annotate band collapses to low frequencies	Raise *Search Hz* above the wind/handling rumble.
Merge refuses or warns	Recordings differ in sampling rate; choose a resampling option or remove the odd one.
Mean Spectra will not draw	All checked recordings must share one sampling rate.
Map or elevation does not load	These need an internet connection; type values manually otherwise.
Export All (Excel) is greyed out	Run Spectral Metrics (Spectral tab) or Compute Metrics (Temporal tab) first.
Interface too small or large	Preferences → Text Size.

14.3 Known limits (version 0.8.0)

- Pulses-only metrics (no peaks) omit temporal excursion and per-peak columns.
- Builds exist for x86_64 Windows and Linux and for Intel and Apple Silicon Macs only.
- The software is under active development; behaviours described as *Provisional* may change.

15. Citing the software and licence

Please cite the paper when you publish results obtained with Rthoptera Desk:

> Rivas, F., Brizio, C., Buzzetti, F. M., & Pijanowski, B. (2025). Rthoptera: Standardised insect bioacoustics in R. *Methods in Ecology and Evolution*, 16(6), 1084–1094. https://doi.org/10.1111/2041-210X.70045

To state which software version you used:

> Rivas, F. (2026). *Rthoptera Desk* (Version 0.8.0) [Computer software]. https://github.com/panchorivasf/Rthoptera-Desk

Work using the **Auto-coded Oscillogram** should also cite:

> Brizio, C. (2023). Colour Enhanced Time/Pressure Envelope (CETPE), a novel on-screen rendering of digital sound. *Rivista Italiana di Acustica*, 48(2), 63–71.

Licence: PolyForm Noncommercial License 1.0.0. Free to use, modify and share for any noncommercial purpose; use by educational institutions, public research organisations, charities, environmental-protection organisations and government institutions counts as noncommercial regardless of funding. Commercial use requires a separate licence. This is a source-available, not an open-source, licence.

Appendix A: Glossary of song structures

Term	Meaning in this program
Envelope	Smoothed amplitude (RMS) curve of the recording over time.
Envelope peak	A local maximum of the envelope that passes the detection thresholds.
Pulse (train)	A group of envelope peaks forming one sound burst; also, in drawn selections, a user-labelled pulse.
Motif / echeme	A group of pulses separated by gaps no longer than Max pulse gap.
Motif sequence	A group of motifs separated by gaps no longer than Max motif gap.
Syllable	In Summarize, a run of consecutive pulses treated as one sound (measured onset of first to offset of last).
TE / DE / PCI	Temporal excursion, dynamic excursion and pulse complexity index (see the Rthoptera paper).
Duty cycle	Proportion of time occupied by sound within a unit.
Floor %	Amplitude-detector threshold as a percentage of the envelope range (an offset above the local noise floor when that is active).
FRB / FRT / FRPT	Frequency Range Bottom / Top and Frequency Range Pressure Threshold (Auto-coded Oscillogram).
Nyquist	Half the sampling rate; the highest frequency that can be represented.
dBFS	Decibels relative to digital full scale (0 dBFS = maximum).

Appendix B: Feedback checklist for testers

Because this manual and the program are both provisional, your notes are very valuable. When something is unclear or wrong, please send:

1. The **version** (Preferences → Version) and your operating system.
2. The tab and the exact buttons you pressed, in order.
3. What you expected and what happened (a screenshot helps; the bottom-right notice text is useful).
4. If possible, a short recording (a few seconds) that reproduces it, and the preset `.json` you used.
5. Any place where this manual disagrees with the program, or where a term was not understood.

Also welcome: suggestions for the sections marked *Provisional* (column dictionary, Habitus, worked examples for your taxa).